Then, we rotate $\mathcal{A}$ so that it lies along the $\hat{n}$ axis. What is the probability of observing $\sigma_n = +1$? According to Principle 4, and using the row and column expansions of $\langle u|$ and $|\lambda_1\rangle$, the answer is

$$P(+1) = |\langle u|\lambda_1\rangle|^2 = \cos^2\frac{\theta}{2}. \qquad (3.24)$$

Similarly, for the same setup,

$$P(-1) = |\langle u|\lambda_2\rangle|^2 = \sin^2\frac{\theta}{2}. \qquad (3.25)$$

With this result, we have come nearly full circle. When introducing spins, we made the claim that if we prepare a large number of them in the *up* state and then measure their component along $\hat{n}$, at angle θ to the z axis, then the average value of the measured results would be $\cos\theta$—the same result we would get for a simple 3-vector in classical physics. Does our mathematical framework give the same result? *It had better!* If a theory disagrees with experiment, it's the theory that has to leave town. Let's see how well our theory holds up so far.

Unfortunately, we need to cheat a little by using an equation that we will not fully explain until the next lecture. This is the equation that tells us how to calculate the average value (also called the expectation value) of a measurement. Here it is:

$$\langle \mathbf{L}\rangle = \sum_i \lambda_i P(\lambda_i). \qquad (3.26)$$

It's worth mentioning that Eq. 3.26 is just a standard formula for an average value. It's not unique to quantum mechanics.

To calculate the expectation value of a measurement corresponding to the operator **L**, we multiply each eigenvalue by its probability, and then sum the